

Pillar 1: GTM Architecture & Version Governance

Container Taxonomy, Event Triggering & Deployment Lifecycle | Apple Store Measurement Setup

 **CONTAINER NAME**
apple-store-container

 **PUBLIC ID**
GTM-K8K5N7RK

 **TARGET ANALYTICS ENGINE**
GA4 ·
G-X3R766JV6K

 **DEPLOYMENT STATUS**
 Demo build

01 Context & Scope

The executive narrative



PROJECT OBJECTIVE

Design an enterprise-level, privacy-compliant tracking architecture for an e-commerce platform (Apple Store Demo), capable of supporting accurate attribution and product-level revenue reporting at scale.

The measurement layer for this program was built on a decoupled, event-driven foundation rather than brittle DOM listeners. Every user interaction of business significance, beginning with `add_to_cart`, is pushed into a structured `dataLayer` object and consumed asynchronously by the container. This removes the tracking logic's dependency on specific CSS selectors or DOM structure, so front-end redesigns don't silently break measurement.



Google Tag Manager

Container & trigger orchestration



Google Analytics 4

Event & ecommerce reporting



Google Ads Conversion Linker

First-party click ID persistence

02 Container Architecture & Tag Taxonomy

How the container is organized, and why each tag fires where it does



GA4 – Base Configuration

GOOGLE TAG

Deployed on the **Initialization - All Pages** hook, ahead of every other tag in the container, to establish the persistent user session and measurement ID before any event-scoped tag can fire.

Fires on `Initialization - All Pages`



Conversion Linker

Also deployed at initialization, this tag persists click identifiers (gclid) into first-party cookie context, preserving attribution integrity under Safari ITP and Firefox ETP constraints without relying on third-party storage.

Fires on **Initialization – All Pages**



GA4 Event – add_to_cart

DECOUPLED TRIGGER

A custom event listener bound directly to the add_to_cart dataLayer push described in Section 4, rather than to a button click or CSS class. The trigger fires purely on the presence of the event in the data layer, independent of front-end markup.

Fires on **Custom Event – add_to_cart**

03 Implementation Proof

Visual assets from the demo container

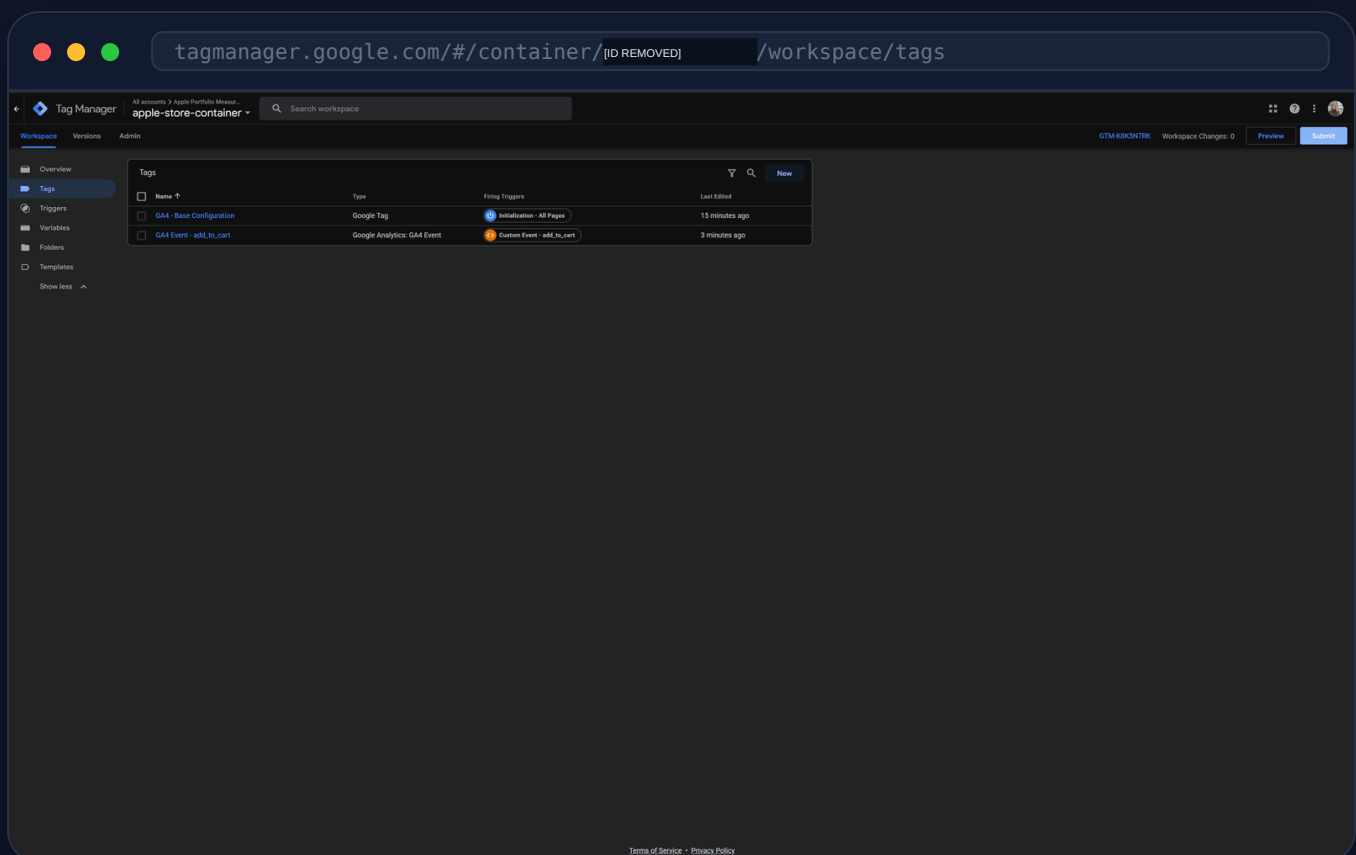


FIG. 1 Container Tag Architecture — demonstrates custom event tagging (GA4 Event – add_to_cart), the base configuration tag, and clean, self-documenting naming conventions across the workspace.

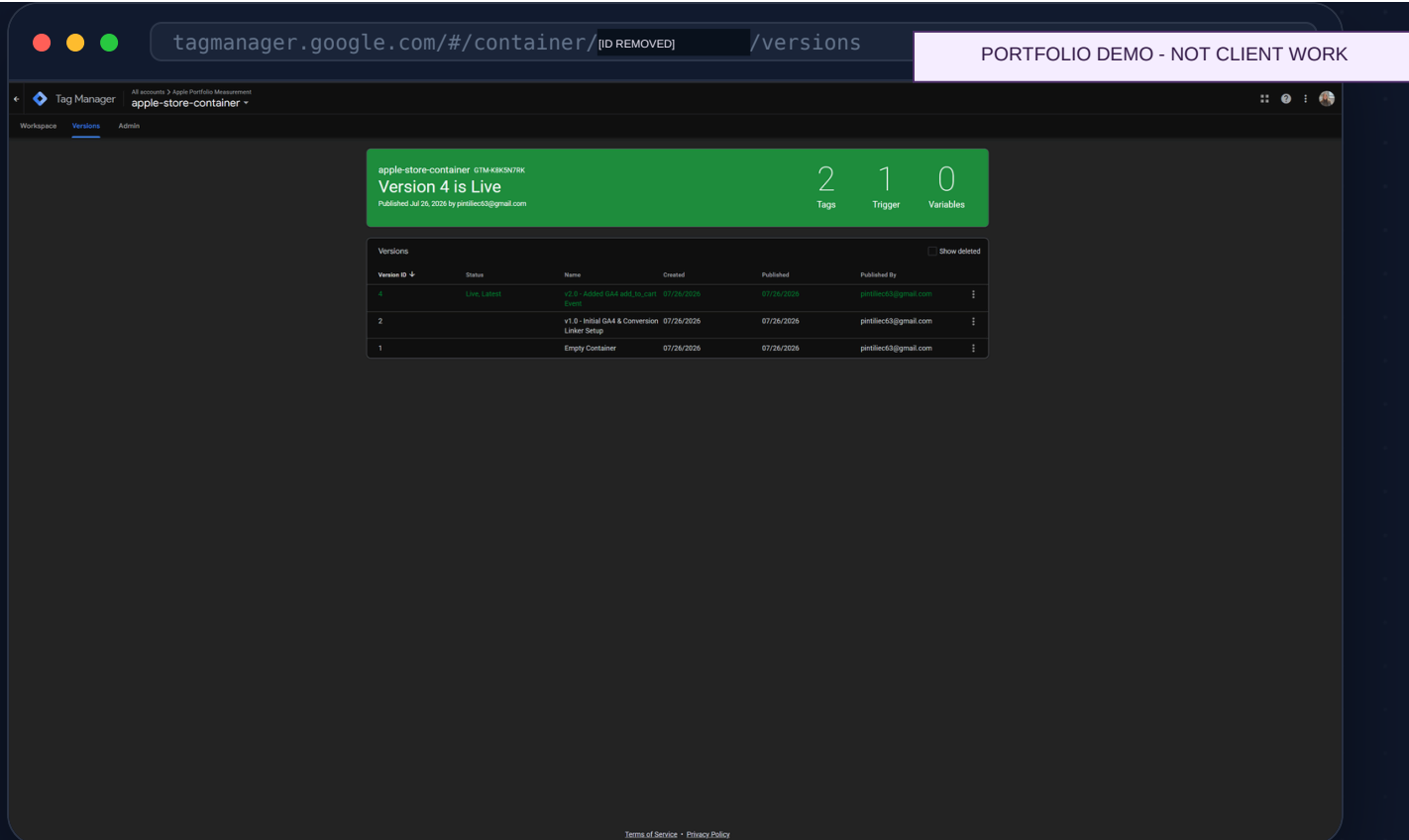


FIG. 2 Container Version Governance — shows disciplined release management across v1.0 and v2.0, with a full audit history and controlled, attributable deployment at every step.

04 The Technical Data Layer Spec

Front-end schema handed to engineering for implementation

The snippet below is the exact contract shared with front-end engineering: the precise shape, keys, and types the `dataLayer` push must match for the `add_to_cart` event to be picked up correctly by the trigger defined in Section 2.

JS ecommerce-add-to-cart.js

dataLayer.push()

```
// Data Layer push when a user adds a product to cart
window.dataLayer = window.dataLayer || [];
window.dataLayer.push({ ecommerce: null }); // Clear previous ecommerce object
window.dataLayer.push({
  event: "add_to_cart",
  ecommerce: {
    currency: "USD",
    value: 1199.00,
    items: [
      {
        item_id: "IPHONE-15-PRO-128",
        item_name: "iPhone 15 Pro 128GB",
        price: 1199.00,
        quantity: 1,
        item_category: "Smartphones"
      }
    ]
  }
});
```